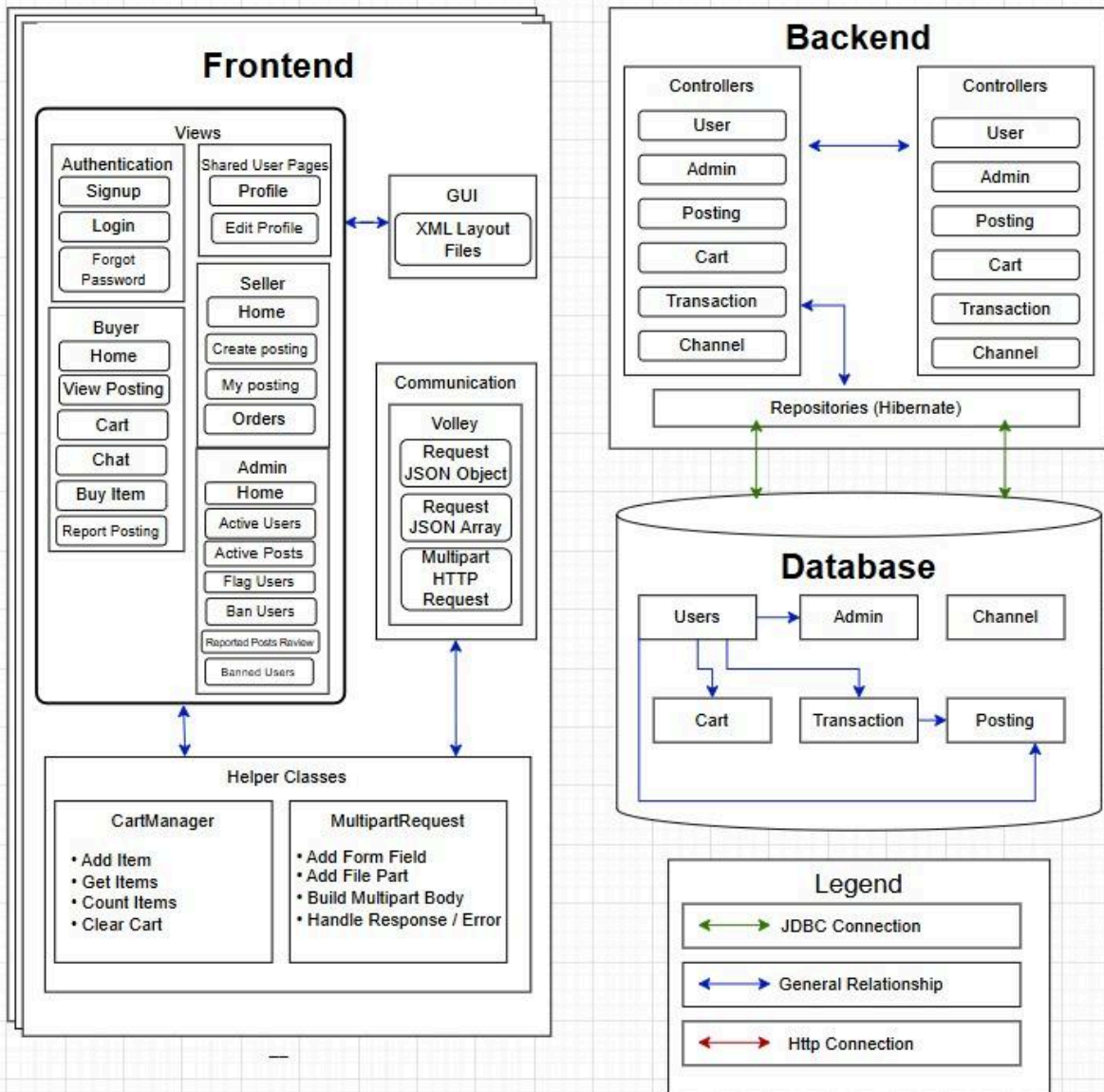

Design Document for **Cy Collectibles**

Group **1_sb_8**

Anish Reddy Nukala: Frontend

Prajwal Reddy Chenreddy: Frontend

Harikrishna Guddati: Backend



Frontend (currently implemented)

LoginActivity (Admin, Buyer, Seller): LoginActivity generates a page with the following elements:

- o EditText: Username
- o EditText: Password
- o Button: Login

When the user clicks 'Login', the app checks the input, sends the login data to the server, stores session data, and redirects the user based on account type.

BuyerHomeActivity (Buyer): BuyerHomeActivity generates a page with the following elements:

- o Button: Category
- o Button: Sort
- o EditText: Search
- o Dynamic posting list

This page is a complex part of the design because it supports searching, sorting, filtering by genre, and dynamically loading postings from the backend.

CartManager: CartManager handles cart-related server communication. It supports:

- o addToCart()
- o getCart()
- o clearCart()
- o removeFromCart()

This design is more complex because cart actions are separated from the activity, making the code easier to organize and reuse.

Backend

Communication

The backend uses REST API mappings to communicate with the frontend:

- Post: create users, postings, channels, and transactions
- Get: retrieve users, postings, notifications, messages
- Patch/Put: update users, postings, and admin actions
- Delete: remove postings or deactivate users

Controllers

- UserController: handles login, signup, user updates, and deletion with validation and conflict checks
- PostingController: handles creating listings, image upload, filtering/searching postings, reporting, and deletion
- TransactionController: processes purchases, updates posting status, removes cart items, and stores transaction history
- NotificationController: retrieves user notifications and marks them as read
- ChannelController: manages messaging (direct/group channels, members, and chat data)
- AdminController: handles banning users/postings, flagging users, and reviewing reported listings

Complex Design

- Purchase flow updates multiple systems (posting, cart, transaction)
- Messaging uses channels + memberships + messages together
- Admin actions can cascade (banning a user bans all their postings)
- Posting system supports dynamic filtering, sorting, and search

Overall, complexity comes from coordinating multiple controllers, database relationships, and backend rules across the system.

